

Literature Status: Abelian WAP Groups

Run `fa.banach.001`

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Status. Literature already answered. This packet records that a question from Megrelishvili, *Fragmentability and representations of flows*, arXiv:math/0411112, has a later negative literature answer.

Source question. In Section “Reflexive representations of (semi)groups” of the source paper (page 13 of the source PDF), after recalling that the additive group of $L_4[0, 1]$ is WAP but not unitarily representable, Megrelishvili asks:

Is every abelian Hausdorff topological group, for example the additive group of a Banach space, WAP?

Here WAP for a topological group means that the universal WAP/semitopological compactification is an embedding, equivalently that WAP functions induce the topology; this is the standard reflexive-representability condition for topological groups.

Supporting answer. Ben Yaacov–Berenstein–Ferri, *Reflexive representability and stable metrics*, arXiv:0901.1003, explicitly recall the Megrelishvili and Glasner–Megrelishvili problem asking whether there exists an abelian group that is not reflexively representable. Their Section 4 then proves:

The additive group of the Banach space c_0 is not reflexively representable.

This is Theorem 4.1 on page 8 of the supporting PDF.

Identification. The additive group of c_0 , with its norm topology, is an abelian Hausdorff topological group. If it were WAP in the sense of the source question, then its topology would be induced by WAP functions, hence it would be reflexively representable. Theorem 4.1 of Ben Yaacov–Berenstein–Ferri says this is impossible. Thus c_0 gives a negative answer to Megrelishvili’s abelian WAP question.

Provenance and search evidence. This is an explicit literature answer, not an original counterexample from the run: the supporting paper states that it is addressing the abelian non-reflexively-representable problem and gives c_0 as the example. The local run indexes were searched for 0411112, 0901.1003, `every abelian Hausdorff`, `c_0 reflexively representable`, and nearby WAP/reflexive-representability keywords; no existing durable packet for this exact source/supporting-paper pair was found.

Scope limitations. This packet only records the negative answer to the abelian Hausdorff WAP question. It does not settle the source paper’s other fragmentability, Asplund-representation, or flow-representation questions.

References

- [1] M. Megrelishvili, *Fragmentability and representations of flows*, arXiv:math/0411112.
- [2] I. Ben Yaacov, A. Berenstein, and S. Ferri, *Reflexive representability and stable metrics*, arXiv:0901.1003; Math. Z. 267 (2011), 129–138.